

ML4HOR: Machine Learning for Health Outcomes with R

Full Program, September 2026 Cohort

Imad EL BADISY, PhD

1 Course Overview

Title: Machine Learning for Health Outcomes with R

Format: Online, cohort-based

Duration: 2 weeks, 8 sessions, 2 hours per session, 16 contact hours

Dates: 21/09/2026 – 04/10/2026

Days: Monday, Wednesday, Friday (18h00 to 20h00), Sunday (10h00 to 12h00)

Time: Morocco time

Instructor: Imad EL BADISY, PhD

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1.1 Course Description

ML4HOR provides a practical introduction to machine learning for health outcomes research using R. The course covers the complete supervised learning workflow across classification, regression, and time-to-event (survival) outcomes, and explores clustering as a final extension for patient phenotyping, subgroup discovery, and exploratory health-data analysis.

The course emphasizes validation, data leakage prevention, appropriate performance metrics (including censoring-aware metrics for survival), calibration, interpretability, clinical relevance, and reproducible reporting.

1.2 Target Audience

Researchers, PhD students, clinicians, public health professionals, and data analysts. Basic R and statistical literacy recommended.

1.3 Prerequisites

Basic familiarity with R (as covered in BIOSTATR) is recommended but not mandatory.

1.4 Learning Objectives

By the end of ML4HOR, participants will be able to:

1. formulate machine learning questions in health outcomes research;
2. prepare tabular health datasets for ML analysis;
3. fit supervised learners for classification, regression, and time-to-event (survival) outcomes;
4. evaluate models using resampling-based validation and censoring-aware metrics;
5. tune and compare learners fairly;
6. assess discrimination, calibration, and clinical usefulness;
7. interpret ML models using global and local explanation methods;
8. apply clustering for patient phenotyping and subgroup discovery;
9. distinguish prediction, clustering, and causal estimation;
10. produce a reproducible ML report with Quarto.

2 Detailed Schedule

Table 1: ML4HOR detailed schedule

No.	Week	Date	Time	Topic	Main Content
1	1	21/09/2026 (Mon)	18h00- 20h00	Introduction to ML for Health Outcomes & R/Quarto Workflow	Prediction vs explanation, outcome types (classification, regression, time-to-event), supervised vs unsupervised learning, leakage risks, data preparation
2	1	23/09/2026 (Wed)	18h00- 20h00	Resampling and Performance Metrics	Train/test split, cross-validation, bootstrap, RMSE, MAE, AUC, F1, log-loss, Brier score, C-index and IPCW metrics for survival
3	1	25/09/2026 (Fri)	18h00- 20h00	Linear and Regularized Models	GLM, logistic regression, penalized Cox, ridge, lasso, elastic net, shrinkage, high-dimensional predictors
4	1	27/09/2026 (Sun)	10h00- 12h00	Tree-Based Models, Random Forests, Boosting & Ensembles	CART, random forests, random survival forests, variable importance, XGBoost, LightGBM, SVM, kNN, stacking
5	2	28/09/2026 (Mon)	18h00- 20h00	Hyperparameter Tuning, Model Selection & Benchmarking	Grid/random search, nested CV, optimism, fair comparison, benchmark tables
6	2	30/09/2026 (Wed)	18h00- 20h00	Calibration and Clinical Utility	Calibration curves, discrimination vs calibration, thresholds, sensitivity, specificity, ROC-AUC
7	2	02/10/2026 (Fri)	18h00- 20h00	Model Interpretation, Explainability & Clustering for Phenotyping	Permutation importance, PDP, ICE, ALE, SHAP, k-means, hierarchical clustering, cluster profiling

No.	Week	Date	Time	Topic	Main Content
8	2	04/10/2026 (Sun)	10h00- 12h00	End-to-End ML Pipeline and Mini-Project	Full workflow: prepare, fit or cluster, evaluate, tune/compare, calibrate, interpret, report

3 Course Materials

- Slides and self-contained notebooks distributed before each session
- R scripts, datasets, and exercises (shared repository)
- Exercises reviewed and corrected during sessions

4 Assessment

Each course ends with a short reproducible Quarto report or mini-project, presented and discussed in Session 8.

5 About the Instructor

Imad EL BADISY, PhD, is a biostatistician working on statistical and machine learning methods for health data, with a focus on survival analysis. Full profile, publications, and R packages: elbadisyimad.com

6 Policies

Language: Teaching in French, with English for statistical and methodological terminology.

Software: R (≥ 4.4) and RStudio required.

Attendance: Full attendance at all 8 sessions is recommended given the cumulative structure.